

# ALOK SINGH KATARIA

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GitHub: <https://github.com/alokkataria101/quantum-physics>

## PHYSICS - COMPUTATIONAL SCIENCE - SOFTWARE ENGINEERING - AI - QUANTUM COMPUTING

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### Profile

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**Physicist and Senior Software Engineer** with 25+ years of professional experience across software development, cloud platforms, DevOps, automation, enterprise systems, and EdTech. **IIT Kanpur Physics graduate** with a strong foundation in quantum mechanics, mathematical physics, linear algebra, probability, and numerical methods.

Hands-on experience with Python, scientific computing, AI/ML, and quantum-computing tools. Independent technical work includes **quantum mechanics, numerical methods, Hamiltonian and eigenvalue problems, VQE, quantum simulation, quantum chemistry, and quantum optimization** using Python and Qiskit.

Combines Physics and mathematical foundations with extensive software-engineering experience to **model, simulate, analyze, and prototype computational solutions**. Also experienced in developing mathematical and educational software and working with AI/ML approaches in EdTech.

**Interested in opportunities involving Physics, computational science, scientific software, EdTech, AI/ML, and practical applications of quantum computing.**

### Core Areas

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Computational Physics - Quantum Mechanics - Mathematical Physics

Scientific Computing - Numerical Methods - Linear Algebra - Optimization

Quantum Computing - Quantum Algorithms - Quantum Simulation

VQE - Quantum Chemistry - Quantum Optimization

Quantum Noise & Error Mitigation

### Selected Quantum & Computational Physics Work

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- Applied classical numerical methods to quantum systems including the **hydrogen atom, 1-D particle-in-a-box, and harmonic oscillator**, using Python, NumPy, SciPy, linear algebra, and the **Finite Difference Method (FDM)** to construct Hamiltonians and compute energy eigenvalues, comparing numerical results with analytical solutions where available.
- For systems without convenient closed-form solutions, explored approximate numerical approaches using Python and **AI-assisted computational methods** for formulation, implementation, and validation.
- Applied a similar computational approach to quantum-computing formulations, using **qubit representations, Hamiltonians, quantum circuits, and VQE** with Python and Qiskit.
- Explored quantum-chemistry workflows using **STO-3G, Hartree-Fock, molecular Hamiltonians, and variational ansatz** for ground-state energy estimation, comparing results with classical calculations.
- Used classical numerical solutions as reference points for exploring corresponding **quantum formulations and variational approaches**, along with quantum optimization and quantum-information concepts such as portfolio optimization, qubit encoding, quantum randomness, spin dynamics, Hadamard transformations, and measurement probabilities.

### Professional Experience

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IBM -- Advisory Software Engineer / DevOps Engineer | 2022 - Present

**IBM Cloud Platform Re-platforming, SRE & Compliance Automation**

Worked on re-platforming and operational engineering for IBM Cloud-based enterprise services, including

Kubernetes/IKS environments, networking, IAM, databases, CI/CD, and security/compliance automation.

- **Private Endpoint & Bastion-based Cluster Access:** Led migration of cluster access from public endpoints/VPN-based access to private-endpoint-only access through IBM Cloud bastion infrastructure, including end-to-end networking and access configuration.
- **Privileged Access Gateway (PAG):** Evaluated and implemented PAG-based access as an alternative to the existing bastion and GPVPN approach, including testing and integration activities.
- **Auditree Compliance Automation:** Configured and integrated Auditree-based compliance checks and evidence collection across IBM Cloud accounts and resources.
- **Compliance Data Processing & Automation:** Developed and maintained Jenkins/Tekton automation pipelines for processing compliance information, including IAM, database, cluster, and infrastructure data, with reporting and remediation workflows.
- **Database Security Scanning:** Automated security scanning and reporting workflows for MongoDB and PostgreSQL databases using enterprise security tooling.
- **Security & Compliance Add-ons:** Worked with EDR, syslog forwarding, vulnerability scanning, and related security controls as part of platform compliance requirements.
- **Cloud / SRE:** Worked across IBM Cloud networking, Kubernetes/IKS, IAM, databases, CI/CD pipelines, security tooling, and production operational automation.

#### Wipro -- Lead Software Developer | 2022

Software development and engineering work across enterprise applications and cloud-based systems.

- Worked on the **CreditMemo** process for the Commissioning System for **Telefónica Germany**.
- Developed services for loading and processing incoming data using Python, Flask, and Pandas.
- Implemented aggregation and validation, feature enhancements, defect fixes, and unit testing.
- Worked with PostgreSQL and Google Cloud.

#### RevenueBox -- Senior Software Engineer | 2021

B2B Account-Based Marketing (ABM) and go-to-market software focused on identifying, prioritizing, and engaging the right customer accounts.

- Developed a Python microservice to identify, process, and select relevant customer accounts based on multiple input signals.
- Designed a mathematical scoring model to calculate and assign a **"Quibble Score"** for account prioritization and tracking, using Python, Docker, MongoDB, and Google Cloud.

#### Lumos Learning -- Research Scientist / Software Engineer | 2020

Online examination and learning platform with emphasis on **mathematics and automated question/solution generation**.

- Developed Python-based mathematical question-generation and solution-processing components, including difficulty estimation and automated evaluation workflows.
- Worked with **AWS Lambda, SQL Server, NLTK, Python, and neural-network/ML approaches** for educational applications.
- Contributed to mathematics-focused chatbot and educational automation initiatives.

#### HCL Technologies -- Senior Software Engineer | 2017 - 2018

L3 product support for **IBM AppScan** (web application security testing), troubleshooting complex customer issues, root-cause analysis, and coordinating fixes across SAST/DAST scanning and reporting workflows.

- Worked on AppScan SAST/DAST capabilities, customer configurations, defect analysis, and production issue resolution.

#### IBM India Software Labs -- Senior Staff Software Engineer | 2008 - 2017

L3 Development & Troubleshooting Engineer for **IBM InfoSphere Data Replication (IIDR)** (previously DataMirror), specializing in complex production issues, root-cause analysis, debugging, and customer escalations.

- Expert in heterogeneous data replication and integration, including **DB2 for i / AS400, Oracle, DB2, SQL Server, and Teradata**, with IBM messaging and remote journaling technologies.
- Strong experience in cross-language and cross-platform integration, including **AS400/IBM i** and COBOL-based enterprise systems; supported customer ETL and data-movement workflows.
- Recognized as a performance specialist, contributing to performance analysis and optimization and receiving an **IBM award for performance expertise**.
- Contributed to feature development, product maintenance, automation, and build/release pipeline engineering across multiple product releases.
- Worked with C, Java, Python, RPG, COBOL, AS400, and Jenkins.

## CA Technologies -- Software Engineer | 2005 - 2008

Worked on **eTrust Access Control, eTrust Audit, and eTrust Single Sign-On** -- enterprise security products.

- Developed more than 100 automated tests for nightly builds and ported product code to AIX.
- Investigated customer and QA defects including memory corruption, enhancements, and installation/upgrade issues.
- Worked with C, Python, Linux, UNIX, AIX, Solaris, and HP-UX.

## Technical Skills

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**Programming:** Python, C, Java, Shell, JavaScript

**Scientific Computing:** NumPy, SciPy, Pandas, MATLAB, scikit-learn, TensorFlow

**Quantum Computing:** Qiskit, VQE, Quantum Circuits, Quantum Simulation

**Cloud & DevOps:** IBM Cloud, AWS, Google Cloud, Kubernetes, OpenShift, Docker, Jenkins, Tekton, CI/CD

**Databases:** PostgreSQL, MongoDB, SQL Server, Oracle, DB2, Teradata

**Operating Systems:** Linux, AIX, Solaris, HP-UX, Windows, IBM i / AS400

## Selected Research & Technical Work

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### Quantum Intuition: From Spin Dynamics to the Hadamard Gate and pi Estimation via Quantum Randomness

Explored quantum randomness and quantum-state transformations through Qiskit-based simulations, including spin dynamics, Hadamard transformations, measurement probabilities, and Monte Carlo estimation of pi.

**Status:** Under IBM internal review for publication.

### Classical VQE Demonstration for the Hydrogen Atom

Developed a classical demonstration of the VQE approach for the hydrogen atom, connecting the variational principle, Hamiltonian formulation, parameterized states, energy expectation values, and numerical optimization.

**Status:** Under IBM internal review for publication.

## Intellectual Property

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- **US Patent #9,528,562** -- Granted US patent related to mechanics.
- Two additional mechanics-related inventions under IBM patent review.
- Internal IBM intellectual-property work related to **quantum portfolio optimization**.

## Education

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### IIT Kanpur -- Integrated M.Sc. in Physics | 2000

Academic foundation includes Quantum Mechanics, Classical Mechanics, Electromagnetism, Mathematical Physics, Linear Algebra, Probability, Differential Equations, Numerical Methods, and Statistical Physics.

## Academic Distinctions & Awards

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- IIT-JEE AIR 1211
- IBM Excellence and Eminence Award -- 2012
- IIT Kanpur alumnus
- Strong academic foundation in Physics and Advanced Mathematics

## Physics Teaching & Mentoring

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IIT-JEE Advanced Physics Faculty / Mentor

*Narayana - PIE Education - Vizag Vikas College*

- Taught advanced IIT-JEE Physics with emphasis on **conceptual understanding, mathematical formulation, and rigorous problem solving**.
- Worked extensively with mechanics, electromagnetism, modern physics, and mathematical approaches to advanced Physics problems.
- Mentored students preparing for highly competitive engineering entrance examinations.

## Research & Code

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Selected work includes computational quantum mechanics, numerical Hamiltonian methods, quantum simulation, VQE, quantum chemistry, quantum optimization, and quantum-information experiments using Python and Qiskit.